# TEST 1—FORM CODE 7C

**SECTION 1** Time—30 minutes 45 Questions

**For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.**

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. GRITTY: (A) smooth and soft (B) cold and silent (C) bright and colorful (D) healthy (E) elongated

2. RESOLVE: (A) willingness (B) indecision (C) dependency (D) reliability (E) irrelevance

3. OPAQUE: (A) impoverished (B) brittle (C) spiked (D) transparent (E) noisy

4. DEGRADATION: (A) saturation (B) elevation (C) perturbation (D) fascination (E) procrastination

5. CRINGE: (A) wait impatiently (B) react quickly (C) treat kindly (D) fold neatly (E) stand boldly

6. UNWARY: (A) alert (B) sad (C) angry (D) eager (E) dutiful

7. HAMPER: (A) ogle (B) clarify (C) assist (D) uncover (E) ventilate

8. DISCREPANCY: (A) expansion (B) correspondence (C) consequence (D) lack of judgment (E) increase in worth

9. GAUNT: (A) rotund (B) garish (C) nervous (D) stylish (E) valiant

10. MORATORIUM: (A) period of activity (B) arrogant behavior (C) picturesque scene (D) successful contestant (E) unexpected response

11. CASTIGATE: (A) collect (B) hesitate (C) salute (D) deteriorate (E) mobilize

12. ABSTEMIOUS: (A) proud (B) overindulgent (C) undervalued (D) easily overtaken (E) having many talents

13. WHET: (A) evade (B) ingest (C) solidify (D) deaden (E) reap

14. ANTEDILUVIAN: (A) homemade (B) jingoistic (C) newfangled (D) well-balanced (E) sweet-scented

15. RIFE: (A) rare (B) weak (C) progressive (D) adaptable (E) enjoyable

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Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate
(C) sophisticated. .amateur (D) risqué. .crude
(E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. As the Jacksons cheerfully poured forth a stream of personal revelations, it soon became evident that ---- was not the distinguishing characteristic of this couple.

(A) volubility (B) discursiveness (C) shyness
(D) enthusiasm (E) gaiety

17. Laboratories have been warned that provisions for animal protection that in the past were merely ---- will now be mandatory; ---- of this policy will lose their federal research grants.

(A) comprehensive. .adversaries
(B) nominal. .advocates
(C) disregarded. .proponents
(D) recommended. .violators
(E) compulsory. .resisters

18. Because the increase in the birth rate during that decade was dramatic and unprecedented, demographers judged that this rise was ---- and that the birth rate would gradually ----, but they were surprised by yet another increase.

(A) characteristic. .flourish
(B) enduring. .rebound
(C) recurrent. .terminate
(D) temporary. .diminish
(E) beneficial. .deviate

19. To Judith, traveling was ----; her sister, however, looked upon each trip as an ---- experience.

(A) confusing. .unnerving
(B) joyous. .exciting
(C) exhilarating. .interminable
(D) stupefying. .unhappy
(E) tiring. .exhausting

20. The film ---- to present a believable portrait of life because it ---- the seemingly insignificant details that give each person's experience its unique nature.

(A) seeks. .deletes (B) refuses. .supplies
(C) manages. .ignores (D) fails. .omits
(E) contrives. .distorts

21. The project's objective is to find and map every remaining example of these medieval earthworks before they are ---- by modern agricultural machinery.

(A) superseded (B) stabilized (C) disclosed
(D) replicated (E) obliterated

22. If society is to make creative use of its internal tensions, we need to see differences as ---- rather than mutually threatening.

(A) illusory (B) ominous (C) coercive
(D) detrimental (E) complementary

23. The revised 1983 edition of Colish's pathfinding study, first published in 1968, incorporates her valuable scholarship of the ---- years and thereby reflects the ---- of the author's thoughts.

(A) culminating. .preparation
(B) intervening. .refinement
(C) following. .stagnation
(D) intermittent. .chronology
(E) final. .immaturity

24. At its best, scientific investigation is a ---- activity; researchers know that their work will be open to general scrutiny, and so they ---- their arguments in defense of their positions.

(A) careful. .neglect (B) secret. .strengthen
(C) tedious. .invent (D) public. .marshal
(E) scholarly. .denounce

25. Novelist Jeffery Paul Chan is committed to ---- the myth of an exotic Chinatown, a myth that he believes has been an impediment to the development of a more positive identity for young Chinese-Americans.

(A) disseminating (B) debunking (C) forging
(D) condoning (E) romanticizing

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Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

An observer watching the sky on any clear moonless night can easily observe streaks of light flashing across it every now and then. The ancients interpreted these flashes as "falling stars," and this name is still used by many people in referring to such phenomena. This colorful description gives an entirely wrong impression of the flashes. They are actually caused by meteors, often not much larger nor more massive than grains of sand, which enter the Earth's atmosphere at a speed of about 20 miles per second. Most meteors are disintegrated by the intense heat generated by friction with the air. The temperature of the meteor and of the air surrounding it increases to a few thousand degrees as the kinetic energy of the meteor is quickly transformed into heat; one then sees a sudden luminous streak across the sky whose length, duration, and brightness depend on the size, mass, and speed of the meteor. If it were not for the deceleration resulting from the Earth's atmosphere, many meteoric particles striking the Earth would have an impact greater than that of a .45-caliber bullet and would be very destructive. The number of meteors that enter the Earth's atmosphere in a single 24-hour period and are bright enough to be seen is extremely large. In fact, these meteors are the source of the several tons of meteoric or cosmic dust that are added to the Earth and its atmosphere every day.

Since the debris that separates from the head of a comet continues to move as a stream of matter in the orbit of the comet, a much larger number of meteors is observed when the Earth passes close to such an orbit. In fact, one then observes what is called a meteor shower, with all the meteors in the shower appearing to diverge from, or converge to, a single point in the sky. This phenomenon is illusory; the meteors are really traveling parallel to each other, but they seem to converge to or diverge from a point in the sky because parallel lines seem to intersect at infinity. During the Leonid shower—one whose apparent convergence point was in the constellation of Leo—that occurred on November 13, 1833, meteors were so abundant that many thousands were seen at some observing stations. Such rich showers, however, are rare. Twelve distinct meteor showers, almost one a month, occur every year, but most of them are not very impressive. Each of these is associated with the orbit of a comet.

26. According to the passage, the intensity of the streak caused by a meteor is influenced by which of the following?

I. The dimensions of the meteor
II. The origin of the meteor
III. The velocity of the meteor

(A) I only
(B) I and II only
(C) I and III only
(D) II and III only
(E) I, II, and III

27. Which of the following statements about the kinetic energy of a meteor traveling through the Earth's atmosphere can be accurately inferred from the passage?

(A) It increases the closer the meteor gets to the Earth's surface.
(B) It serves to neutralize the atmosphere's effect on the meteor.
(C) It is partially converted into light.
(D) It is not sufficient to make the meteor visible.
(E) It cannot be precisely calculated.

28. According to the author, a meteor shower results when

(A) the Earth passes near the path of a comet
(B) a massive comet begins to disintegrate
(C) meteor clusters collide and break up
(D) the kinetic energy of a meteor is transformed into heat
(E) meteors converge to a single point in the sky

29. With which of the following statements concerning meteors would the author be most likely to agree?

(A) Meteors can alter the surface of a planet without an atmosphere.
(B) Meteors diverge from a single point in the sky.
(C) Meteors converge to a single point in the sky.
(D) The speed of a meteor determines its mass.
(E) The brightness of a meteor indicates its distance from Earth.

30. The author does all of the following in developing the passage EXCEPT

(A) refute a mistaken notion
(B) use an illustrative analogy
(C) explain an optical illusion
(D) refer to a historical event
(E) present a personal anecdote

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The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

In certain non-Western societies, a scholar once suggested, the institution of communal music "gives to individuals a solid center in an existence that seems to be almost chaos, and a continuity in their being that would otherwise too easily dissolve before the calls of the implacable present. Through its words, people who might be tempted to give in to the malice of circumstances find their old powers revived or new powers stirring in them, and through these life itself is sustained." This, I think, sums up the role played by song in the lives of Black American slaves. Songs of the years before Emancipation supply abundant evidence that in the structure of the music, in the survival of oral tradition, and in the ways the slaves expressed their Christianity, important elements of their common African heritage became vitally creative aspects of their Black American culture.

Although it was once thought that Africans newly arrived in America were passive recipients of European cultural values that supplanted their own, most African slaves in fact were so isolated from the larger American society beyond the plantation that European cultural influence was no more than partial. Through necessity they drew on and reestablished the only cultural frame of reference that made any sense to them, that of cultures in which they had been raised, and thus passed on distinctly African cultural patterns to slaves born and raised in America. One example of the process of cultural adjustment is the response to Christianity, valued by American-born slaves not as an institution imported intact from another culture but as a spiritual perception of heroic figures and demonstration of divine justice that they could use to transcend the bonds of their condition through the culturally significant medium of song. Earlier historians frequently failed to perceive the full importance of this because they did not take seriously enough the strength of feeling represented in the sacred songs. A religion that was a mere anodyne imposed by an oppressing culture could never have inspired a music as forceful and striking to all who witnessed it as the religious music of the American slaves. Historians who have tried to argue that these people did not oppose the institution of slavery in any meaningful collective way reason from a narrowly twentieth-century Western viewpoint. Within the frame of reference inherited from African cultures in which the functions of song included criticism and mockery of rulers, there were meaningful methods of opposition to authority and self-assertion so foreign to most Western historians as to be unrecognizable. Modern Americans raised in a wholly Western culture need to move mentally outside their own culture, in which music plays only a peripheral role, before they can understand how American-born slaves put to use the functions music had had for their African ancestors.

31. Which of the following statements best expresses the main idea of the passage?

(A) Communal song was a vital part of the heritage of American-born slaves.
(B) Communal music was primarily important as a recreational pastime.
(C) Communal song had several functions in ancient African cultures.
(D) Non-Western cultures give Western historians insights into elements of their own culture.
(E) Songs prized by American slaves reminded them of their legendary homelands.

32. Underlying the scholar's description of communal music (lines 2-9) is the assumption that human existence

(A) presents a rich variety of subjects for popular songs
(B) sometimes reveals profound artistic truths
(C) often seems bewildering and hopeless
(D) crushes the spirit of even the most resourceful artist
(E) rarely encourages artistic expression

33. The passage suggests that the "strength of feeling" (line 37) expressed in the slaves' religious music was a direct reflection of which of the following?

(A) The use of religion to distract people from present problems
(B) The sense of injustice as inevitable and inescapable
(C) The perception of Christianity as a distinctly foreign institution
(D) The role of song as a vehicle for social commentary
(E) The importance of music to Western society

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34. The sentence that begins "A religion that was . . ." (lines 38-42) makes all of the following points EXCEPT:

(A) Christianity had important spiritual significance for Black American slaves.
(B) People who have assumed that slaves considered Christianity only a superficial, alien institution are mistaken.
(C) The power of the slaves' religious music indicates how deeply they felt what they sang.
(D) The slaves' religious music provided a moving experience for the listeners as well as the singers.
(E) The religious music of American-born slaves made only indirect use of Christian figures.

35. The author states that historians have sometimes misunderstood the music of American slaves for which of the following reasons?

I. They could not personally observe the music being sung by American slaves.
II. Some historians interpreted the significance of this music within the wrong cultural context.
III. Many Western historians felt uncomfortable with the presentation of religious stories through art.

(A) I only
(B) II only
(C) III only
(D) I and II
(E) I and III

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN:BOREDOM :: (A) dream:sleep (B) anger:madness (C) smile:amusement (D) face:expression (E) impatience:rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. LEOTARD : DANCER ::
(A) apron : chef (B) badge : detective (C) baton : conductor (D) hammer : carpenter (E) microscope : scientist

37. SHRUG : SHOULDERS :: (A) grin : jokes (B) hug : friends (C) frown : lines (D) bite : food (E) nod : head

38. TRUNK : AUTOMOBILE :: (A) pilot : airplane (B) passenger : train (C) elevator : building (D) closet : house (E) gear : transmission

39. BASKETBALL : BASKET :: (A) football : foot (B) soccer : goal (C) baseball : stadium (D) bowling : gutter (E) tennis : racket

40. EPIC : LITERATURE ::
(A) plot : fiction
(B) etching : frame
(C) symphony : music
(D) blueprint : construction
(E) accompaniment : ballet

41. INTEREST : OBSESSION ::
(A) confidence : conceit (B) gasp : terror (C) intelligence : talent (D) trust : suspicion (E) punishment : crime

42. CORRODED : METAL :: (A) diseased : antibody (B) erased : mistake (C) intended : action (D) implied : opinion (E) corrupted : integrity

43. DEPOSE : RULER :: (A) unfrock : priest (B) assassinate : politician (C) delegate : emissary (D) renounce : authority (E) challenge : rival

44. PALISADE : STAKES :: (A) dike : ocean (B) lawn : grass (C) moat : castle (D) hedge : shrubs (E) garden : plants

45. EGOIST : ALTRUISM ::
(A) sluggard : energy (B) antagonist : pain (C) glutton : food (D) masochist : pleasure (E) philanthropist : generosity

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

**SECTION 2** Time—30 minutes, 25 Questions

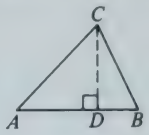
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius $r$: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

| | |
|---|---|
| $=$ is equal to | $\leqq$ is less than or equal to |
| $\neq$ is unequal to | $\geqq$ is greater than or equal to |
| $<$ is less than | $\parallel$ is parallel to |
| $>$ is greater than | $\perp$ is perpendicular to |




Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If $n + n + n = n + 10$, then $n =$

(A) 5
(B) 10
(C) 15
(D) 20
(E) 30

| Skirt Size | Skirt Length |
|---|---|
| 2 | $20\frac{1}{2}$ inches |
| 4 | 22 inches |
| 6 | $x$ inches |
| 8 | 25 inches |
| 10 | $26\frac{1}{2}$ inches |
| 12 | 28 inches |

2. In the table above, if the difference in skirt lengths between any skirt size and the next larger size is constant, what is the value of $x$?

(A) $22\frac{1}{2}$
(B) 23
(C) $23\frac{1}{2}$
(D) 24
(E) $24\frac{1}{2}$

3. Widgets are packed exactly $\frac{1}{2}$ dozen to a carton, and extras are left for the next packing. If a batch of 103 widgets is ready to be packed and if there are enough cartons, how many widgets will be left for the next packing?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

4. If $a$, $b$, and $c$ are integers, the expression $a(b + c)$ is always equal to each of the following EXCEPT

(A) $(a + b)c$
(B) $a(c + b)$
(C) $(b + c)a$
(D) $(c + b)a$
(E) $ab + ac$

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5. If $2y = 3$, then $3(2y)^2 =$

(A) $\frac{27}{4}$

(B) 18

(C) $\frac{81}{4}$

(D) 27

(E) 81

6. Which of the following is equal to $0.00238 \times 10^2$?

(A) 2.38
(B) 0.238
(C) 0.0238
(D) 0.000238
(E) 0.0000238

Questions 7-8 refer to the following definition.

$\begin{matrix} a & b \\ & c \end{matrix} = \frac{a \cdot b}{c} + \frac{b \cdot c}{a} + \frac{c \cdot a}{b}$ for all nonzero $a$, $b$, and $c$.

For example,

$\begin{matrix} 2 & 4 \\ & 6 \end{matrix} = \frac{2 \cdot 4}{6} + \frac{4 \cdot 6}{2} + \frac{6 \cdot 2}{4} = \frac{4}{3} + 12 + 3 = 16\frac{1}{3}$

7. $\begin{matrix} 3 & 12 \\ & 4 \end{matrix} =$

(A) 1
(B) 9
(C) 10
(D) 16
(E) 26

8. If $x \neq 0$, $\begin{matrix} x & x^2 \\ & x^3 \end{matrix} =$

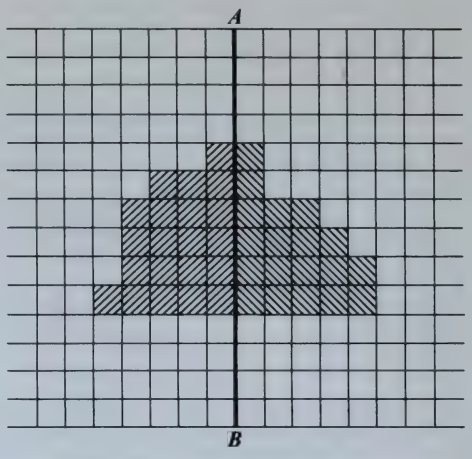
(A) $x^6 + x^4 + x^2$

(B) $x^5 + x + \frac{1}{x}$

(C) $x^4 + x^3 + 1$

(D) $x^4 + x^2 + 1$

(E) $x^2 + x + 1$




9. What is the least number of additional squares that must be shaded in the figure above to make the shaded region symmetric about the line $AB$?

(A) One
(B) Two
(C) Three
(D) Four
(E) Five

10. The number $n - 3$ is how much less than $n + 3$?

(A) 3
(B) 6
(C) $n - 6$
(D) $n - 3$
(E) $2n$

11. How many odd positive integers are less than 36?

(A) 16
(B) 17
(C) 18
(D) 19
(E) 35

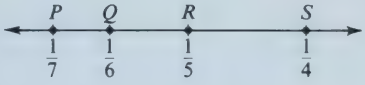
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12. The ratio of the lengths of the sides of two squares is 1:3. What is the ratio of their areas?

(A) 1:9
(B) 1:6
(C) 1:4
(D) 1:3
(E) 2:3

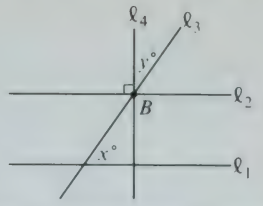
13. If Joe picks strawberries at an average rate of $m$ quarts per hour, approximately how many quarts does he pick in $h$ hours, in terms of $m$ and $h$?

(A) $\frac{m}{h}$
(B) $\frac{h}{m}$
(C) $m - h$
(D) $m + h$
(E) $mh$




14. On the number line above, where does the number 0.13 lie?

(A) To the right of point $S$
(B) Between points $R$ and $S$
(C) Between points $Q$ and $R$
(D) Between points $P$ and $Q$
(E) To the left of point $P$




Note: Figure not drawn to scale.

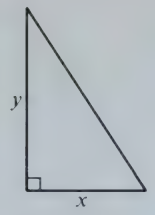
15. In the figure above, line $\ell_2$ is parallel to line $\ell_1$ and is perpendicular to $\ell_4$. $B$ is a point on lines $\ell_2$, $\ell_3$, and $\ell_4$. If $x = y$, what is the value of $x$?

(A) 15
(B) 30
(C) 45
(D) 50
(E) 60

16. Of seven consecutive integers in increasing order, if the sum of the first three integers is 33, what is the sum of the last three integers?

(A) 36
(B) 39
(C) 42
(D) 45
(E) 48

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Note: Figure not drawn to scale.

17. In the right triangle above, if $x$ and $y$ are integers and $x + y = 8$, what is the value of $x$ for which the area of the triangle is maximum?

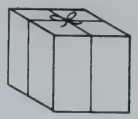
(A) 1
(B) 2
(C) 4
(D) 6
(E) 7

18. In a 3-drawer chest, Ira has 30 percent of his shirts in drawer $A$, 25 percent in drawer $B$, and the rest in drawer $C$. What is the fewest number of shirts he could have in drawer $C$?

(A) 1
(B) 4
(C) 9
(D) 10
(E) 20

19. If $af = 6$, $fg = 1$, $ag = 24$, and $a > 0$, then $afg =$

(A) $\frac{3}{2}$
(B) 4
(C) 6
(D) 12
(E) 144



20. In the figure above, a cubic box with edge of length $x$ inches is tied with a string 106 inches long. The string crosses itself at right angles on the top and bottom of the box. If the bow required 10 inches of string, what is the maximum number of inches $x$ could be?

(A) 12
(B) 13
(C) 16
(D) 17
(E) 18

21. If $x$, $\frac{1}{x}$, $y$, $\frac{1}{y}$, $z$, and $\frac{1}{z}$ are integers, which of the following could NOT be a value of $x + y + z$?

(A) 4
(B) 3
(C) 1
(D) $-1$
(E) $-3$

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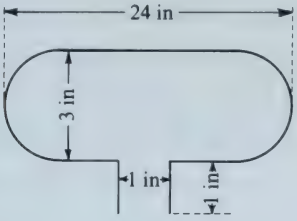
22. If $x - y$ is positive, which of the following could be true about $x$ and $y$?

I. $y < x < 0$

II. $0 < y < x$

III. $x < y < 0$

(A) I only
(B) II only
(C) III only
(D) I and II
(E) I and III




Note: Figure not drawn to scale.

23. The figure above represents a television antenna made completely of wire. The rounded sections of the antenna are semicircles. How many inches of wire are in the part of the antenna shown?

(A) $40 + 3\pi$
(B) $43 + 3\pi$
(C) $43 + 6\pi$
(D) $44 + 3\pi$
(E) $48 + 3\pi$

24. In January Ms. Walker spent $\frac{1}{4}$ of the money in her savings account. The following month she spent $\frac{1}{5}$ of the remainder. If she then had \$3,000 left, how much was in her savings account originally?

(A) \$4,500
(B) \$5,000
(C) \$5,500
(D) \$6,000
(E) \$7,500

25. In $\triangle ABC$, $D$ is the midpoint of $AC$, and another point, $E$, is on $AC$ such that $BE \perp AC$. Which of the following must be true?

(A) $\angle ABD = \angle DBC$

(B) $\angle EBD = \frac{1}{4} \angle ABC$

(C) $\angle BAD = \angle BDA$

(D) The length of $AC >$ the length of $BC$

(E) The length of $BD >$ the length of $BE$

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

**SECTION 3** Time—30 minutes 50 Questions

The questions in this section measure skills that are important to writing well. In particular, they test your ability to recognize and use language that is clear, effective, and correct according to the requirements of standard written English, the kind of English found in most college textbooks.

Directions: The following sentences contain problems in grammar, usage, diction (choice of words), and idiom.

Some sentences are correct.
No sentence contains more than one error.

You will find that the error, if there is one, is underlined and lettered. Assume that elements of the sentence that are not underlined are correct and cannot be changed. In choosing answers, follow the requirements of standard written English.

If there is an error, select the one underlined part that must be changed to make the sentence correct and fill in the corresponding oval on your answer sheet.

If there is no error, fill in oval (E).

EXAMPLE:

The region has a climate so severe that (A) plants growing there (B) rarely had been (C) more than twelve inches high (D). No error (E)

SAMPLE ANSWER

(A) (B) ● (D) (E)

1. A few scientists claim that food additives not only (A) improve the quality of (B) foods but also made (C) them (D) safer. No error (E)

2. It is rumored that (A) the names of them (B) to be promoted (C) will be announced tomorrow, but I believe the choices have not been made (D) yet. No error (E)

3. In 1957, the appearance of the first (A) Soviet satellite has created (B) a panic in the United States that continued for (C) nearly (D) a decade. No error (E)

4. In appreciation about (A) her work, the committee presented its (B) retiring director with, among other (C) gifts, a plaque describing (D) her accomplishments. No error (E)

5. For many people (A), hang-gliding, an increasingly (B) popular sport, seems satisfying (C) the urge to fly (D). No error (E)

6. Despite (A) much research, there are (B) still certain elements in (C) the life cycle of the cicada that are not fully understood (D). No error (E)

7. Louise's fair skin was (A) sunburned so badly (B) that she looked as if (C) she had fell into (D) a bucket of red paint. No error (E)

8. Whenever (A) we hear of a natural disaster, even in (B) a distant part of the world, you (C) feel sympathy (D) for the people affected. No error (E)

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9. Late in the war, the Germans, retreating (A) in haste (B), left many (C) of their (D) prisoners go free. No error (E)

10. In many states (A), there seems (B) to be a belief, openly (C), expressed by educators, that the methods of teaching reading should be changed (D). No error (E)

11. Throughout (A) the Middle Ages women work (B) beside (C) men, knowing that the efforts of men and women alike were essential to (D) survival. No error (E)

12. Without hardly a doubt (A), the novels of Thomas Pynchon are (B) more complex (C) than the novels of many other (D) contemporary writers. No error (E)

13. The leading roles in the widely acclaimed (A) play, a modern (B) version of an Irish folktale, were performed (C) by Jessica and he (D). No error (E)

14. In the reserve section of the library, there is (A) two volumes of (B) essays by James Baldwin on (C) the relation of the Black artist and intellectual to (D) society. No error (E)

15. Many studies have tried to (A) determine whether or not (B) seeing violence on television makes (C) children behave more violent (D). No error (E)

16. Between ten and (A) twenty per cent of the textbook appears to be (B) new; the rest (C) is a revision of (D) the previous edition. No error (E)

17. Some people seem remarkably (A) insensitive to physical pain, and (B) this insensitivity does not mean (C) that they are able (D) to endure other kinds of pain. No error (E)

18. During the meeting (A), Congresswoman Barbara Jordan stressed that (B) educators and legislators must cooperate where (C) the goal of equal opportunity is (D) to be reached. No error (E)

19. The population of American alligators, dangerously (A) small a few years ago, are now estimated at (B) (C) more than one million (D). No error (E)

20. Such (A) novels as *Heidi* and *Little Women* have long been considered (B) by young and old alike (C) to be a classic of (D) children's literature. No error (E)

21. Anyone who (A) gathers mushrooms for the purpose of eating them (B) must distinguish carefully (C) between (D) poisonous and nonpoisonous species. No error (E)

22. In regards to (A) the energy crisis, the President urged (B) all homeowners to keep (C) their (D) thermostats at sixty-five degrees in winter. No error (E)

23. Long ago, well before (A) the invention of (B) the printing press, poets often sung (C) their (D) poetry to small, interested audiences. No error (E)

24. Because of (A) extreme weather conditions, starvation exists (B) in some countries where they (C) must struggle every day to stay alive (D). No error (E)

25. The energy question, along with several other (A) issues, are going to be discussed (B) (C) at (D) the next meeting of the state legislature. No error (E)

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Directions: In each of the following sentences, some part or all of the sentence is underlined. Below each sentence you will find five ways of phrasing the underlined part. Select the answer that produces the most effective sentence, one that is clear and exact, without awkwardness or ambiguity, and fill in the corresponding oval on your answer sheet. In choosing answers, follow the requirements of standard written English. Choose the answer that best expresses the meaning of the original sentence.

Answer (A) is always the same as the underlined part. Choose answer (A) if you think the original sentence needs no revision.

EXAMPLE:

Laura Ingalls Wilder published her first book and she was sixty-five years old then.

(A) and she was sixty-five years old then
(B) when she was sixty-five years old
(C) at age sixty-five years old
(D) upon reaching sixty-five years
(E) at the time when she was sixty-five

SAMPLE ANSWER



26. Freud's complex theory based on the death instinct is, for most people, one that is with difficult understanding.

(A) with difficult understanding
(B) difficult to understand
(C) to understand difficultly
(D) having difficulty being understood
(E) difficult for understanding

27. In the nineteenth century, trains were more than machines they were expressions of the greatness of the United States.

(A) machines they were
(B) machines and were
(C) machines; they were
(D) machines, although they were
(E) machines, but were

28. In the sunlight, the cherry blossoms that burst out everywhere, like foam on breaking waves.

(A) blossoms that burst
(B) blossoms bursting
(C) blossoms, which are bursting
(D) blossoms burst
(E) blossoms, which burst

29. When Dorothy Richardson decided to become a novelist, she knew that her writing would leave her little time for other work.

(A) When Dorothy Richardson decided to become
(B) After the decision was made by Dorothy Richardson to become
(C) After the decision by Dorothy Richardson to become
(D) When Dorothy Richardson decides to become
(E) When Dorothy Richardson decided about becoming

30. The difference between the twins is that one is humorous; the other, serious.

(A) one is humorous; the other, serious
(B) of one being humorous, the other is serious
(C) one is humorous; the other being serious
(D) one is humorous, although the other is more serious
(E) of a humorous one and one that is serious

31. The United States did not go on the gold standard until 1900, it went off it thirty-three years later.

(A) until 1900, it went off it thirty-three years later
(B) until 1900; however, going off after thirty-three years
(C) until 1900, although going off in thirty-three years
(D) until 1900, it was thirty-three years later when it went off it
(E) until 1900 and went off it thirty-three years later

32. Many memos were issued by the director of the agency that had an insulting tone, according to the staff members.

(A) Many memos were issued by the director of the agency that
(B) Many memos were issued by the director of the agency who
(C) The issuance of many memos by the director of the agency which
(D) The director of the agency issued many memos that
(E) The director of the agency, who issued many memos that

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33. The widespread slaughter of buffalo for their skins profoundly shocked and angered Native Americans.
(A) shocked and angered Native Americans
(B) shocked Native Americans, angering them
(C) shocked Native Americans, and they were also angry
(D) was a shock and caused anger among Native Americans
(E) was shocking to Native Americans, making them angry

34. Consumers are beginning to take notice of electric cars because they are quiet, cause no air pollution, and gasoline is not used.
(A) cause no air pollution, and gasoline is not used
(B) air pollution is not caused, and gasoline is not used
(C) cause no air pollution, and use no gasoline
(D) causing no air pollution and using no gasoline
(E) air pollution is not caused, and no gasoline is used

35. Because she was a woman was why Sharon Frontiero, a lieutenant in the United States Air Force, felt that she was being treated unfairly.
(A) Because she was a woman was why Sharon Frontiero, a lieutenant in the United States Air Force, felt that she was being treated unfairly.
(B) Sharon Frontiero, a lieutenant in the United States Air Force, felt that she was being treated unfairly because she was a woman.
(C) Because she was a woman, Sharon Frontiero felt that this was why she was being treated unfairly as a lieutenant in the United States Air Force.
(D) Sharon Frontiero, a lieutenant in the United States Air Force, feeling that she was being treated unfairly because she was a woman.
(E) A woman, Sharon Frontiero, felt that because she was a lieutenant in the United States Air Force, that she was being treated unfairly.

36. A major difference between people and apes is brain size, a person's brain is three times as large as the brain of an ape.
(A) size, a person's brain is
(B) size, with a person's brain being
(C) size; a person's brain is
(D) size; a person's brain, it is
(E) size, in that it is

37. The process of how European immigrant groups being absorbed into American society is complex.
(A) of how European immigrant groups being absorbed
(B) for European immigrant groups being absorbed
(C) where European immigrant groups were absorbed
(D) by which European immigrant groups have been absorbed
(E) whereby the absorption of European groups has been

38. Mr. Howe's class has organized a special program for our school; the purpose being to help us increase our understanding of Japanese culture.
(A) school; the purpose being to
(B) school and the purpose is to
(C) school, the purpose is to
(D) school, being to
(E) school to

39. Light reaching earth from the most distant stars originated billions of years ago.
(A) reaching earth from the most distant stars
(B) which reaching earth from the most distant stars
(C) from the most distant stars reaching earth
(D) that is from the most distant stars and reaches earth
(E) reaching earth which is from stars that are most distant

40. Joseph Conrad was born and educated in Poland and he wrote all of his novels in English.
(A) Joseph Conrad was born and educated in Poland and he
(B) Joseph Conrad, being born and educated in Poland,
(C) Although being born and educated in Poland, Joseph Conrad
(D) Although Joseph Conrad was born and educated in Poland, he
(E) Being from Poland, where he was born and educated, Joseph Conrad

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Note: The remaining questions are like those at the beginning of the section.

Directions: For each sentence in which you find an error, select the one underlined part that must be changed to make the sentence correct and fill in the corresponding oval on your answer sheet.

If there is no error, fill in oval (E).

EXAMPLE:

The region has a climate so severe that (A) plants growing there (B) rarely had been (C) more than twelve inches high. (D) No error (E)

SAMPLE ANSWER

(A) (B) ● (D) (E)

41. If you find that it is (A) (B) difficult to concentrate in noisy surroundings, (C) one should (D) try to find a quiet place to study. No error (E)

42. The speaker claimed that no other modern nation devotes (A) so small a portion of (B) its wealth (C) to public assistance and health as (D) the United States does. No error (E)

43. The thought of trying to (A) persuade their (B) three-year-old to sit in (C) a high chair did not appeal to either the mother nor (D) the father. No error (E)

44. The bright (A) fiberglass sculptures of Luis Jiménez has received (B) critical acclaim not only (C) in his home state, New Mexico, but also in (D) New York. No error (E)

45. Doctors see a connection between (A) increased (B) amounts of leisure time spent sunbathing and (C) the increased number of (D) cases of skin cancer. No error (E)

46. Whether or not (A) credit card companies should prevent their (B) customers to acquire (C) substantial debts was the issue discussed at (D) the meeting. No error (E)

47. The board's final recommendations included (A) hiring additional (B) personnel, dismissing the head of (C) research, and a reorganized marketing (D) division. No error (E)

48. Some people prefer (A) attending movies to television (B) because they (C) dislike the frequent (D) interruptions of programs for commercials. No error (E)

49. Like (A) many factory workers of (B) a century ago, women today (C) are developing organizations to represent (D) their interests. No error (E)

50. In a prominent city newspaper, they claim (A) that the number of (B) unregistered participants in (C) this year's marathon was half that of (D) last year's. No error (E)

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

**SECTION 5** Time—30 minutes 40 Questions

For each question in this section, choose the best answer and fill in the corresponding oval on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. UNCORK: (A) reform (B) empty (C) seal (D) dry out (E) blend in

2. FRAUDULENCE: (A) truthfulness (B) prosperity (C) melancholy (D) composure (E) potency

3. WITHSTAND: (A) learn from (B) copy from (C) add to (D) show to (E) yield to

4. ARCHAIC: (A) minor (B) modern (C) flattened (D) unique (E) urban

5. WISPY: (A) substantial (B) frigid (C) conventional (D) colorful (E) odorless

6. MYRIAD: (A) personal taste (B) accurate image (C) loud group (D) small number (E) nearby place

7. PONTIFICATE: (A) idolize (B) persist (C) speak modestly (D) investigate promptly (E) drive carelessly

8. NEFARIOUS: (A) adroit (B) dapper (C) serious (D) virtuous (E) eloquent

9. BILLOWING: (A) drooping (B) liberated (C) distant (D) silent (E) enthusiastic

10. MAGNANIMOUS: (A) dainty (B) latent (C) vengeful (D) conspicuous (E) compatible

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that, when inserted in the sentence, best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Because of improvements made in the microscope, organisms that heretofore were ---- are now easily studied.

(A) unavoidable (B) indistinguishable (C) inactive (D) intolerable (E) inconsequential

12. I found that the writer's ideas were sufficiently ---- to make me bear with his ---- language.

(A) intriguing. .skill with
(B) interesting. .abuses of
(C) humble. .mastery of
(D) shallow. .errors in
(E) misguided. .style of

13. Paradoxically, this successful entrepreneur is sometimes ---- and at other times reclusive.

(A) autonomous (B) dispassionate (C) solitary (D) unthinking (E) gregarious

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14. Rice has been referred to as a nearly perfect food, in that one can ---- with rice as the major part of one's diet without any ---- physical effects.

(A) survive. .salutary
(B) languish. .harmful
(C) subsist. .deleterious
(D) decline. .deteriorating
(E) thrive. .benign

15. The speaker ---- his initial indiscriminate generalizations by ---- that each segment of the Hispanic population in the United States is a distinct group with its own identity and values.

(A) renounced. .denying
(B) reiterated. .assuming
(C) qualified. .emphasizing
(D) minimized. .doubting
(E) verified. .insisting

**Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.**

**Example:**

**YAWN:BOREDOM :: (A) dream:sleep (B) anger:madness (C) smile:amusement (D) face:expression (E) impatience:rebellion**

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. HEIGHT : MOUNTAIN :: (A) ray : sun (B) length : width (C) peak : valley (D) depth : sea (E) weight : scale

17. RING : BOXER :: (A) grandstand : racehorse (B) stage : actor (C) track meet : runner (D) box office : ticket (E) airport : plane

18. STUDY : KNOWLEDGE ::
(A) accomplishment : incentive
(B) exercise : strength
(C) reality : wish
(D) cookbook : food
(E) time : growth

19. SAND : HOURGLASS :: (A) liquid : bottle (B) strap : wristwatch (C) battery : flashlight (D) line : ruler (E) mercury : thermometer

20. STATUE : SCULPTOR ::
(A) fire : firefighter (B) paint : painter (C) medicine : doctor (D) law : lawyer (E) suit : tailor

21. LAMENTATION : SORROW ::
(A) instigation : plot
(B) reassurance : anxiety
(C) exclamation : surprise
(D) resistance : futility
(E) interruption : occurrence

22. ACQUIRE : STEAL :: (A) capture : interrogate (B) impress : influence (C) testify : perjure (D) denounce : insult (E) frighten : tremble

23. EXCERPT : BOOK :: (A) swatch : fabric (B) revision : manuscript (C) review : film (D) fringe : shawl (E) chapter : index

24. SUPPLANT : REPLACEMENT ::
(A) terrorize : protection
(B) obliterate : destruction
(C) implant : extraction
(D) infiltrate : discovery
(E) supplement : inadequacy

25. MAELSTROM : WATER :: (A) blizzard : wind (B) rock slide : earth (C) tornado : air (D) tidal wave : ocean (E) plateau : land

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Each passage below is followed by questions based on its content. Answer the questions following each passage on the basis of what is stated or implied in that passage.

As soon as cable service was restored after the earthquake, Baron Okura replied to architect Frank Lloyd Wright's inquiry with a message of congratulation: HOTEL STANDS UNDAMAGED AS MONUMENT OF YOUR GENIUS. HUNDREDS OF HOMELESS PROVIDED FOR BY PERFECTLY MAINTAINED SERVICE. CONGRATULATIONS, OKURA.

Never one to display undue reticence in such matters, Wright speedily convened a press conference at which he said nothing to dissuade reporters from drawing the inference that the Imperial Hotel was the only building in Tokyo that had remained standing through the disaster. In fact, however, hundreds of other solid masonry buildings in both Tokyo and Yokohama also withstood the quake—most notably those of British architect Josiah Condor, whose numerous structures suffered considerably less damage than Wright's. Nonetheless, the Imperial Hotel's thoroughly undeserved fame as the only building that had stood up through the great Tokyo quake was to prove far more unshakable than the edifice itself; and Wright's renown as the man who had designed and built it flourished accordingly. While by no means wholly responsible for the architectural revolution that was to revitalize the world's cities during the next four decades, the worldwide repute of Wright's Imperial Hotel was to facilitate and hasten its progress. By the time this famous edifice was demolished in 1967, the great earthquake had been instrumental in altering not only the appearance of Tokyo but also that of many of the other great cities in the world.

26. The primary purpose of the passage is to

(A) outline some of Wright's failings as an architect
(B) describe the architectural revolution brought about by Wright
(C) explain how Wright increased his fame and helped change the world's architecture
(D) attack Wright's architectural abilities and his greed for world renown
(E) compare the historical significance of Wright's work to that of Josiah Condor

27. The passage suggests that Josiah Condor was not as famous as Wright because Condor

(A) was not an innovative architect
(B) did not take advantage of publicity
(C) did not design buildings especially to withstand earthquakes
(D) did not realize the extent of the damages caused by the Tokyo quake
(E) had stopped designing new buildings by the time the architectural revolution took place

28. In developing the passage, the author does all of the following EXCEPT

(A) offer a comparison
(B) make use of historical facts
(C) refer to a written document
(D) rely on recollected private experience
(E) point out a cause and effect relationship

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# 5

In any part of the world where there are pronounced seasonal changes in climate, organisms appear and disappear at particular times with uncanny precision. Biologists have long wondered how they do it, and the answer to this question is even now not entirely clear.

As the Earth spins through its seasonal cycle, several environmental parameters change on an annual basis. Temperatures fluctuate, periods of high and low precipitation alternate, and day length increases and decreases. Like temperature, the time and amount of precipitation in most parts of the world is highly unpredictable on a week-to-week basis, even though seasonal averages may not fluctuate significantly. Unlike temperature and precipitation, however, day length repeats itself with monotonous precision year in and year out; it is not particularly surprising, therefore, that many organisms respond to this parameter.

Response to day length, or photoperiodism, among most groups of higher organisms is well known. However, the most detailed studies have been done with flowering plants and insects—partially because they are readily handled in large numbers under laboratory conditions, and partially because many of them have sufficiently short life cycles that their response to photoperiod is quickly evident and hence accessible to experimentation. In insects the most carefully studied phenomenon is entry into diapause, a dormant state that occurs in different insects at very different developmental stages, from the egg through adulthood. In flowering plants the phenomenon subjected to closest analysis has been the transformation from vegetative to reproductive growth—the initiation and development of flowers—although it has been demonstrated that other phenomena, such as the onset of autumn coloration and entrance into winter dormancy, are also responses to day length.

29. The passage is best described as which of the following?

    (A) A response to a controversial theory of ecological relationships
    (B) An explanation of a newly discovered cycle of growth
    (C) An account of climatic events in different parts of the world
    (D) A list of observations made during the course of one year
    (E) An inquiry into possible causes of certain biological phenomena

30. The author cites all of the following as responses to photoperiodism EXCEPT

    (A) insect entry into diapause
    (B) development of flowers
    (C) short life cycles of plants and insects
    (D) onset of winter dormancy in plants
    (E) start of autumn coloration in leaves

31. According to the passage, the phenomenon of diapause is accurately characterized as which of the following?

    I. A state of inactivity
    II. A response to a change in the number of daylight hours
    III. An effect produced primarily through artificial laboratory conditions

    (A) I only
    (B) II only
    (C) III only
    (D) I and II only
    (E) I, II, and III

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Isabel had never met a person having less of that fault which is the principal obstacle to friendship—the air of reproducing the more tiresome, the stale, the too-familiar parts of one's own character. The gates of Isabel's confidence were opened wider than they had ever been; she said things to this amiable auditor that she had not yet said to anyone. Sometimes she took alarm at her candor: it was as if she had given to a comparative stranger the key to her cabinet of jewels. Afterwards, however, she always remembered that one should never regret a generous error and that if Madame Merle had not the merits she attributed to her, so much the worse for Madame Merle.

There are many amiable people in the world, and Madame Merle was far from being vulgarly good-natured and restlessly witty. She knew how to think and she had thought to very good purpose. Of course, too, she knew how to feel; Isabel couldn't have spent a week with her without being sure of that. This was indeed Madame Merle's most perfect gift. Life had told upon her; it was part of the satisfaction to be taken in her society that when Isabel talked of what she was pleased to call serious matters Madame Merle understood her so easily and quickly. Emotion, it is true, had become with Madame Merle rather historic; she made no secret of the fact that the fount of passion, thanks to having been rather violently tapped at one period, didn't flow quite so freely as of yore.

32. The primary purpose of this passage is to

(A) contrast Madame Merle's worldliness with Isabel's naïveté
(B) investigate Madame Merle's idiosyncratic character
(C) illustrate by example the ideal friendship between women
(D) describe the reasons for Isabel's attraction to Madame Merle
(E) emphasize Madame Merle's coarseness by describing Isabel's sophistication

33. It can be inferred from the passage that the "generous error" in line 11 refers to Isabel's

(A) willingness to believe Madame Merle worthy of her confidence
(B) careless freedom with her personal jewelry
(C) reluctance to be critical of Madame Merle's behavior
(D) acceptance of Madame Merle as an equal
(E) desire to give Madame Merle financial assistance

34. The author's attitude toward "many amiable people" (line 14) is best expressed by which of the following?

(A) They represent a majority of the population.
(B) They are boisterous but sensitive and able to think to good purpose.
(C) They are prone to be high-spirited more often than they are amiable.
(D) They deserve the praise and admiration of others.
(E) They are often forced and excessive in their friendliness.

35. With which of the following observations concerning friendship would Isabel most likely agree?

(A) Sympathetic understanding is not absolutely essential to a strong friendship.
(B) A friendship is difficult with a person who echoes the wearisome aspects of one's own character.
(C) One must avoid expressing confidences to new friends.
(D) People should never reveal more of their feelings to friends than those friends share in return.
(E) One should always be careful not to attribute merits to a friend before they have been proven.

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# 5

(This passage was written in 1960.)

Where government has entered directly into the field of art, the experience has too often been disheartening. Political influences have exerted themselves. The standards of the artist and the critic have been ignored or, where these have been permitted a measure of authority, the tendency has been for an artistic clique to entrench itself. The art that has been encouraged under official auspices has almost always favored the less adventurous and the more classically hidebound schools.

From this experience, leading figures in the art world have drawn the conclusion that anything is better than the intrusion of government. It may be questioned, however, whether such individuals in the United States are not thinking too narrowly as professionals, without an adequate understanding of the governmental methods and institutions that in other fields, no less delicate than art, have permitted our political system to act with detachment and a regard for the highest and most sophisticated standards.

The functions performed by the National Institute of Health and the National Science Foundation are, to take two modern examples, quite as complex as the encouragement of art; they are performed with a high level of competence and professional skill. Certainly we should be capable of devoting as much ingenuity to the creation of adequate processes for the nurturing of art as we have for promoting health or science.

It may be argued that, although public officials may not think of themselves as experts on health or science, they may assume they know what is good in art. There is a need for widespread acceptance of the idea that art has its own standards, irrespective of one's personal preferences; that it is professional, disciplined, and subject to the rational and informed judgments of the critics. Unless these principles are accepted, it is perhaps impossible to hope that government can promote art without debasing it. But even that is not reason to despair. There already exist precedents of high officials upholding the verdicts of juries whose views on art differed admittedly from their own. There are also notable examples of government cultural activity successfully protected from political pressure. The Freer Gallery and the National Gallery of Art remind us how private and public efforts can combine to produce results of highest excellence.

36. The passage as a whole is concerned primarily with which of the following questions?

(A) Why is the government interested in promoting art?
(B) Should the government have a role in supporting the arts?
(C) How much influence does the government have over artistic standards?
(D) Why have artists not been more involved in government-sponsored art projects?
(E) How have political influences affected artistic standards in the United States?

37. The author indicates that the art sponsored by governments has tended to be

(A) unpopular (B) grandiose (C) conventional (D) unrealistic (E) exotic

38. Why does the author criticize those art experts who believe the government should not enter into the field of art?

(A) They do not know enough about the successful functioning of governmental institutions in other fields.
(B) They have disapproved of art that was encouraged under official auspices.
(C) They have established impossible standards for government-sponsored art.
(D) They have placed too much trust in politicians' ability to evaluate art.
(E) They are reluctant to speak out in defense of artistic standards.

39. The author mentions the National Gallery of Art chiefly as an example of

(A) successful government involvement with art
(B) government cultural activity affected by political pressure
(C) an institution similar in organization to the National Institute of Health
(D) the entrenchment of an artistic clique in a government project
(E) an institution uninfluenced by the opinions of experts

40. Which of the following does the author include among the "principles" mentioned in line 35?

(A) Only an artist can properly evaluate his or her own work.
(B) The government can support art without devaluating artistic standards.
(C) There are critical standards by which judgments about art can be made.
(D) The appreciation of art is a matter of subjective preference.
(E) Public officials can best determine the proper allocation for government funds in the field of art.

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

**SECTION 6** Time—30 minutes, 35 Questions

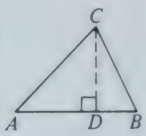
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius $r$: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

| | |
|---|---|
| $=$ is equal to | $\leqq$ is less than or equal to |
| $\neq$ is unequal to | $\geqq$ is greater than or equal to |
| $<$ is less than | $\parallel$ is parallel to |
| $>$ is greater than | $\perp$ is perpendicular to |




Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures that accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

1. If the ratio of $a$ to $b$ is $\frac{7}{3}$, then the ratio of $2a$ to $b$ is

(A) $\frac{7}{6}$
(B) 2
(C) $\frac{7}{3}$
(D) 3
(E) $\frac{14}{3}$

2. On any given day, 8:30 a.m. is how many minutes past 7:15 a.m. ?

(A) 15
(B) 45
(C) 75
(D) 105
(E) 115

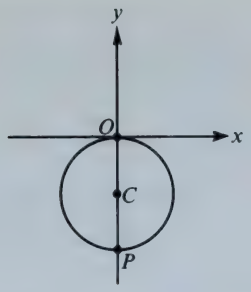
3. If $x + y = 8$ and $xy = 15$, then $x - y$ could equal

(A) 2
(B) 4
(C) 6
(D) 7
(E) 9

4. $\frac{1}{5} - \frac{2}{10} + \frac{3}{15} - \frac{4}{20} =$

(A) $-\frac{2}{5}$
(B) $-\frac{1}{5}$
(C) 0
(D) $\frac{2}{5}$
(E) $\frac{1}{2}$

GO ON TO THE NEXT PAGE



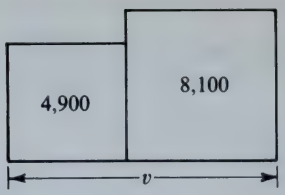

5. In the figure above, the circle with center $C$ has radius 5 and passes through $O$. The coordinates $(x, y)$ of point $P$ are

(A) $(-10, 0)$
(B) $(-5, 0)$
(C) $(-5, 5)$
(D) $(0, -5)$
(E) $(0, -10)$

$$\begin{array}{r} 3\ 7 \\ \times\ 2\ F \\ \hline H\ 4\ G \\ 7\ 4 \\ \hline 8\ 8\ G \end{array}$$

6. If $F$, $G$, and $H$ each represent a different digit in the correctly worked multiplication problem above, what digit does $F$ represent?

(A) 3
(B) 4
(C) 5
(D) 6
(E) 7




7. The figure above is composed of two squares with areas as shown. What is the value of $v$?

(A) 60
(B) 65
(C) 130
(D) 160
(E) 650

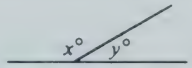
GO ON TO THE NEXT PAGE

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet fill in oval

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

**AN E RESPONSE WILL NOT BE SCORED.**

EXAMPLES

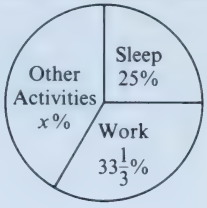
| | Column A | Column B | Answers |
|---|---|---|---|
| E1. | $2 \times 6$ | $2 + 6$ | ● Ⓑ Ⓒ Ⓓ Ⓔ |
| E2. |  $180 - x$ | $y$ | Ⓐ Ⓑ ● Ⓓ Ⓔ |
| E3. | $p - q$ | $q - p$ | Ⓐ Ⓑ Ⓒ ● Ⓔ |

Notes:

1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as $x$, $n$, and $k$ stand for real numbers.

| | Column A | Column B |
|---|---|---|
| 8. | $1 + 11 + 111 + 1{,}111$ | $9 + 99 + 999$ |
| 9. | $\frac{20}{35}$ | $\frac{40}{70}$ |
| | $x = 24 - y$<br>$y - 6 = 7$ | |
| 10. | $x$ | $y$ |
| | $50\%$ of $z = 90$ | |
| 11. | $z$ | 45 |

DISTRIBUTION OF TIME FOR ACTIVITIES OVER A 24-HOUR PERIOD




| | | |
|---|---|---|
| 12. | The number of hours available for "Other Activities" | 12 hours |

| | Column A | Column B |
|---|---|---|
| |  | |

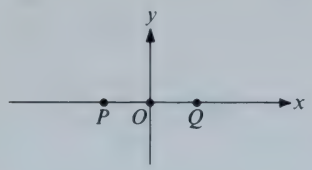
A "tripath" is a path along exactly three connected line segments from ○ to any point marked ●.

| | | |
|---|---|---|
| 13. | The number of different tripaths possible | 9 |
| | $x > 0$ | |
| 14. | The average (arithmetic mean) of $x$, $2x$, and 30 | The average (arithmetic mean) of $x$ and $2x$ |

GO ON TO THE NEXT PAGE

Column A | Column B

Questions 15-16 refer to the following graph.




15. $x$-coordinate of point $P$ | $x$-coordinate of point $Q$

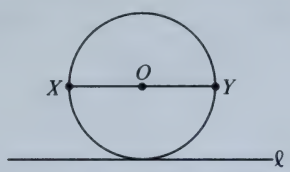
16. $y$-coordinate of point $P$ | $y$-coordinate of point $Q$

---

$2 + r < s$

17. $r$ | $s$

---




Line $\ell$ is tangent to the circle with center $O$.
$XY = 9$

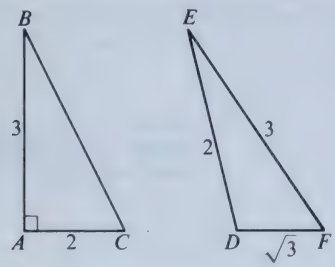
18. The distance from $O$ to any point on $\ell$ | 4

---

$m > 0$
$n > 0$

19. $m + n$ | $mn$

---




Note: Figures not drawn to scale.

20. Perimeter of $\triangle ABC$ | Perimeter of $\triangle DEF$

Column A | Column B

Three line segments with a common end point form three angles of measures $x°$, $y°$, and $z°$.

21. The value of $x + y$ | 180

---

$p < 0$

22. $p^2 + p^3 + p^4 + p^5$ | 0

---

$(x + 5)(x - 3) = 0$

23. $x$ | 5

---

$\frac{6}{25} + \frac{6}{25} + \frac{6}{25} = \frac{x}{75}$

24. $x$ | 36

---

An isosceles right triangle has area 18.

25. The length of its hypotenuse | $6\sqrt{2}$

---

A bicyclist traveled 19 kilometers in $n$ hours at an average speed of 32 kilometers per hour.

26. $n$ | $\frac{2}{3}$

---

Of 100 freshmen, 30 are taking history but not calculus and 50 are taking calculus but not history.

27. The number of freshmen taking both history and calculus | 19

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and fill in the corresponding oval on the answer sheet.

28. The sum of two consecutive positive integers is never divisible by

(A) 2
(B) 3
(C) 5
(D) 7
(E) 9

29. Equal amounts of peanuts, cashews, and almonds are packed separately in bags. If 3 bags, one of each kind, cost a total of \$3.00 and 2 bags of peanuts and 2 bags of cashews cost a total of \$3.50, how much does 1 bag of almonds cost?

(A) \$1.25
(B) \$1.50
(C) \$1.75
(D) \$2.00
(E) \$2.25

30. If $A$, $B$, $C$, and $D$ are four distinct points on a circle such that the four chords $AB$, $BC$, $CD$, and $DA$ are of equal length, the diameter of the circle is equal to the length of which of the following?

(A) $AD$

(B) $AC$

(C) $CD$

(D) $\frac{AD + BC}{2}$

(E) $\frac{AB + BD}{2}$

31. If $\frac{x}{3} = x^2$, the value of $x$ can be which of the following?

I. $-\frac{1}{3}$

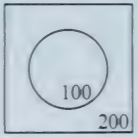
II. $0$

III. $\frac{1}{3}$

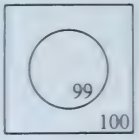
(A) I only
(B) II only
(C) III only
(D) II and III only
(E) I, II, and III

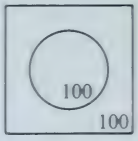


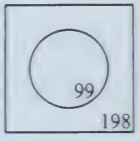

32. Jean uses the symbol above to indicate that a circle of radius $X$ can be drawn in the interior of a square with side of length $Y$ without touching the sides of the square. Which of the following is correct?

(A) 


(B) 


(C) 

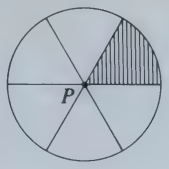

(D) 


(E) 


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33. If the average (arithmetic mean) of six numbers is $-6$, and the sum of four of the numbers is 20, what is the average of the other two numbers?

(A) 7
(B) 8
(C) $-8$
(D) $-28$
(E) $-32$




34. The circle above with center $P$ is divided into 6 equal regions. If the area of the circle is $36\pi$, what is the perimeter of the shaded region?

(A) $12\pi + 6$
(B) $6\pi + 12$
(C) $6\pi + 6$
(D) $2\pi + 36$
(E) $2\pi + 12$

35. A 2-cup mixture consists of $\frac{1}{3}$ flour and $\frac{2}{3}$ cornmeal. If 1 cup of flour is added to make a 3-cup mixture, approximately what percent of the 3-cup mixture is flour?

(A) 65%
(B) 55%
(C) 50%
(D) 45%
(E) 35%

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT TURN TO ANY OTHER SECTION IN THE TEST. **STOP**

# How to Score the Sample Test

Before you can find out what your scores are on the College Board 200 to 800 scale, you need to determine your verbal, mathematical, and TSWE raw scores. The steps for doing so for each section of the test and a scoring worksheet are provided below and on page 66. Use the table on page 67 to determine your correct and incorrect answers for each section.

## Determining Your Raw Scores

### SAT-Verbal Sections 1 and 5

**Step A:** Count the number of correct answers for section 1 and record the number in the space provided on the worksheet. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

**Step B:** Count the number of correct answers and the number of incorrect answers for section 5 and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

**Step C:** To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

**Step D:** To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

### SAT-Mathematical Sections 2 and 6

**Step A:** Count the number of correct answers and the number of incorrect answers for section 2 and record the numbers in the spaces provided on the worksheet. To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

**Step B:** Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35)* in section 6 and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

**Step C:** Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27)* in section 6 and record the numbers in the spaces provided on the worksheet. To determine subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

**Note: Do not count any E responses to questions 8 through 27 as correct or incorrect. Because these four-choice questions have no E answer choices, E responses to these questions are treated as omits.**

**Step D:** To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

**Step E:** To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

### TSWE: Section 3

**Step A:** Count the number of correct answers for section 3 and record the number in the space provided on the worksheet. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine your unrounded raw score, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{total unrounded raw score}$$

**Step B:** To obtain B, your raw TSWE score, round A to the nearest whole number. (For example, any number from 34.50 to 35.49 rounds to 35.) Enter the resulting figure on the worksheet.

# SCORING WORKSHEET FOR THE SAMPLE TEST

## SAT-Verbal Sections

A. Section 1: _____ no. correct $-\frac{1}{4}$ (_____ no. incorrect) = _____ subtotal A

B. Section 5: _____ no. correct $-\frac{1}{4}$ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw verbal score (Rounded to nearest whole number) _____ D

## SAT-Mathematical Sections

A. Section 2: _____ no. correct $-\frac{1}{4}$ (_____ no. incorrect) = _____ subtotal A

B. Section 6: Questions 1 through 7 and 28 through 35 (5-choice) _____ no. correct $-\frac{1}{4}$ (_____ no. incorrect) = _____ subtotal B

C. Section 6: Questions 8 through 27 (4-choice) _____ no. correct $-\frac{1}{3}$ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw math score (Rounded to nearest whole number) _____ E

## TSWE

A. Section 3: Total unrounded raw score _____ no. correct $-\frac{1}{4}$ (_____ no. incorrect) = _____ A

B. Total rounded raw TSWE score (Rounded to nearest whole number) _____ B

# ANSWERS TO SAMPLE TEST QUESTIONS AND PERCENTAGE OF STUDENTS ANSWERING EACH QUESTION CORRECTLY

| Section 1—Verbal | | | Section 2—Mathematical | | | Section 3—TSWE | | | Section 5—Verbal | | | Section 6—Mathematical | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Question number | Correct answer | Percentage of students answering the question correctly | Question number | Correct answer | Percentage of students answering the question correctly | Question number | Correct answer | Percentage of students answering the question correctly | Question number | Correct answer | Percentage of students answering the question correctly | Question number | Correct answer | Percentage of students answering the question correctly |
| 1 | A | 93% | 1 | A | 91% | 1 | C | 86% | 1 | C | 95% | 1 | E | 93% |
| 2 | B | 89 | 2 | C | 89 | 2 | B | 91 | 2 | A | 88 | 2 | C | 84 |
| 3 | D | 77 | 3 | A | 86 | 3 | B | 81 | 3 | E | 77 | 3 | A | 89 |
| 4 | B | 68 | 4 | A | 79 | 4 | A | 84 | 4 | B | 63 | 4 | C | 80 |
| 5 | E | 64 | 5 | D | 76 | 5 | C | 87 | 5 | A | 42 | 5 | E | 66 |
| 6 | A | 66 | 6 | B | 79 | 6 | E | 90 | 6 | D | 33 | 6 | B | 78 |
| 7 | C | 53 | 7 | E | 82 | 7 | D | 77 | 7 | C | 26 | 7 | D | 64 |
| 8 | B | 34 | 8 | D | 59 | 8 | C | 78 | 8 | D | 24 | 8 | A | 96 |
| 9 | A | 29 | 9 | D | 65 | 9 | C | 71 | 9 | A | 36 | 9 | C | 84 |
| 10 | A | 36 | 10 | B | 72 | 10 | E | 78 | 10 | C | 11 | 10 | B | 70 |
| 11 | C | 22 | 11 | C | 73 | 11 | B | 82 | 11 | B | 94 | 11 | A | 69 |
| 12 | B | 28 | 12 | A | 66 | 12 | A | 74 | 12 | B | 61 | 12 | B | 67 |
| 13 | D | 23 | 13 | E | 56 | 13 | D | 73 | 13 | E | 56 | 13 | A | 68 |
| 14 | C | 23 | 14 | E | 62 | 14 | A | 91 | 14 | C | 28 | 14 | D | 37 |
| 15 | A | 7 | 15 | C | 59 | 15 | D | 50 | 15 | C | 21 | 15 | B | 59 |
| 16 | C | 81 | 16 | D | 45 | 16 | E | 68 | 16 | D | 91 | 16 | C | 73 |
| 17 | D | 82 | 17 | C | 45 | 17 | B | 60 | 17 | B | 89 | 17 | B | 67 |
| 18 | D | 94 | 18 | C | 28 | 18 | C | 57 | 18 | B | 92 | 18 | A | 63 |
| 19 | C | 74 | 19 | D | 23 | 19 | B | 65 | 19 | E | 86 | 19 | D | 57 |
| 20 | D | 76 | 20 | A | 36 | 20 | D | 52 | 20 | E | 84 | 20 | A | 68 |
| 21 | E | 70 | 21 | A | 22 | 21 | E | 79 | 21 | C | 54 | 21 | D | 45 |
| 22 | E | 47 | 22 | D | 24 | 22 | A | 44 | 22 | C | 33 | 22 | D | 21 |
| 23 | B | 40 | 23 | B | 15 | 23 | C | 62 | 23 | A | 40 | 23 | B | 65 |
| 24 | D | 44 | 24 | B | 19 | 24 | C | 39 | 24 | B | 35 | 24 | A | 47 |
| 25 | B | 24 | 25 | E | 18 | 25 | B | 67 | 25 | C | 32 | 25 | C | 29 |
| 26 | C | 72 | | | | 26 | B | 94 | 26 | C | 56 | 26 | B | 37 |
| 27 | C | 43 | | | | 27 | C | 68 | 27 | B | 67 | 27 | D | 21 |
| 28 | A | 56 | | | | 28 | D | 83 | 28 | D | 70 | 28 | A | 52 |
| 29 | A | 56 | | | | 29 | A | 79 | 29 | E | 61 | 29 | A | 40 |
| 30 | E | 71 | | | | 30 | A | 50 | 30 | C | 41 | 30 | B | 38 |
| 31 | A | 51 | | | | 31 | E | 65 | 31 | D | 40 | 31 | D | 26 |
| 32 | C | 32 | | | | 32 | D | 63 | 32 | D | 45 | 32 | B | 34 |
| 33 | D | 23 | | | | 33 | A | 82 | 33 | A | 53 | 33 | D | 16 |
| 34 | E | 32 | | | | 34 | C | 69 | 34 | E | 24 | 34 | E | 17 |
| 35 | B | 30 | | | | 35 | B | 90 | 35 | B | 38 | 35 | B | 12 |
| 36 | A | 85 | | | | 36 | C | 54 | 36 | B | 34 | | | |
| 37 | E | 89 | | | | 37 | D | 70 | 37 | C | 36 | | | |
| 38 | D | 79 | | | | 38 | E | 40 | 38 | A | 31 | | | |
| 39 | B | 74 | | | | 39 | A | 60 | 39 | A | 51 | | | |
| 40 | C | 59 | | | | 40 | D | 44 | 40 | C | 25 | | | |
| 41 | A | 53 | | | | 41 | D | 76 | | | | | | |
| 42 | E | 39 | | | | 42 | E | 70 | | | | | | |
| 43 | A | 21 | | | | 43 | D | 68 | | | | | | |
| 44 | D | 10 | | | | 44 | B | 59 | | | | | | |
| 45 | A | 11 | | | | 45 | E | 59 | | | | | | |
| | | | | | | 46 | C | 53 | | | | | | |
| | | | | | | 47 | D | 62 | | | | | | |
| | | | | | | 48 | B | 50 | | | | | | |
| | | | | | | 49 | E | 46 | | | | | | |
| | | | | | | 50 | A | 45 | | | | | | |

**Notes:** The percentages for the SAT-verbal and SAT-mathematical sections are based on the analysis of the answer sheets for a random sample of juniors and seniors who took this test in May 1987 and whose mean scores were 435 on the SAT-verbal sections and 483 on the SAT-mathematical sections.

The percentages for TSWE are based on the analysis of the answer sheets for a random sample of all students who took this test in June 1981 and whose mean score was 43.

## SCORE CONVERSION TABLE
## Sample SAT and TSWE

| | College Board Scaled Score | | | College Board Scaled Score | | | College Board Scaled Score |
|---|---|---|---|---|---|---|---|
| Raw Score | SAT-Verbal | SAT-Math | Raw Score | SAT-Verbal | SAT-Math | Raw Score | TSWE |
| 85 | 800 | | 40 | 460 | 600 | | |
| 84 | 780 | | 39 | 450 | 590 | | |
| 83 | 760 | | 38 | 450 | 580 | | |
| 82 | 750 | | 37 | 440 | 570 | | |
| 81 | 740 | | 36 | 430 | 560 | 50 | 60+ |
| 80 | 730 | | 35 | 430 | 550 | 49 | 60+ |
| 79 | 720 | | 34 | 420 | 540 | 48 | 60+ |
| 78 | 720 | | 33 | 410 | 530 | 47 | 60+ |
| 77 | 710 | | 32 | 410 | 520 | 46 | 59 |
| 76 | 700 | | 31 | 400 | 510 | 45 | 58 |
| 75 | 690 | | 30 | 390 | 510 | 44 | 57 |
| 74 | 690 | | 29 | 390 | 500 | 43 | 56 |
| 73 | 680 | | 28 | 380 | 490 | 42 | 55 |
| 72 | 670 | | 27 | 370 | 480 | 41 | 54 |
| 71 | 660 | | 26 | 370 | 470 | 40 | 53 |
| 70 | 660 | | 25 | 360 | 460 | 39 | 52 |
| 69 | 650 | | 24 | 350 | 450 | 38 | 51 |
| 68 | 640 | | 23 | 340 | 440 | 37 | 50 |
| 67 | 630 | | 22 | 340 | 430 | 36 | 49 |
| 66 | 630 | | 21 | 330 | 420 | 35 | 48 |
| 65 | 620 | | 20 | 320 | 410 | 34 | 47 |
| 64 | 610 | | 19 | 310 | 400 | 33 | 46 |
| 63 | 600 | | 18 | 300 | 390 | 32 | 45 |
| 62 | 600 | | 17 | 300 | 380 | 31 | 44 |
| 61 | 590 | | 16 | 290 | 370 | 30 | 43 |
| 60 | 580 | 800 | 15 | 280 | 370 | 29 | 42 |
| 59 | 580 | 780 | 14 | 270 | 360 | 28 | 41 |
| 58 | 570 | 760 | 13 | 270 | 350 | 27 | 40 |
| 57 | 560 | 750 | 12 | 260 | 340 | 26 | 39 |
| 56 | 560 | 740 | 11 | 250 | 330 | 25 | 38 |
| 55 | 550 | 730 | 10 | 250 | 320 | 24 | 37 |
| 54 | 540 | 720 | 9 | 240 | 310 | 23 | 36 |
| 53 | 540 | 710 | 8 | 230 | 300 | 22 | 35 |
| 52 | 530 | 700 | 7 | 230 | 300 | 21 | 34 |
| 51 | 530 | 690 | 6 | 220 | 290 | 20 | 33 |
| 50 | 520 | 680 | 5 | 220 | 280 | 19 | 32 |
| 49 | 510 | 670 | 4 | 210 | 270 | 18 | 31 |
| 48 | 510 | 660 | 3 | 200 | 260 | 17 | 30 |
| 47 | 500 | 660 | 2 | 200 | 260 | 16 | 29 |
| 46 | 490 | 650 | 1 | 200 | 250 | 15 | 28 |
| 45 | 490 | 640 | 0 | 200 | 240 | 14 | 27 |
| 44 | 480 | 630 | −1 | 200 | 230 | 13 | 26 |
| 43 | 480 | 620 | −2 | 200 | 220 | 12 | 25 |
| 42 | 470 | 610 | −3 | 200 | 210 | 11 | 24 |
| 41 | 460 | 600 | −4 or below | 200 | 200 | 10 | 22 |
| | | | | | | 9 | 21 |
| | | | | | | 8 or below | 20 |